QR Code Tracking (PTZ)

1. Experiment Objective

The PTZ pan-tilt tracks the position of a QR code while decoding its content and displaying it in the terminal and on the frame.

2. Experiment Principle

1. Key Terms

Term	Description
QR Code	A two-dimensional matrix barcode that can encode data such as URLs and text; decoded with the pyzbar or OpenCV library
pyzbar	A Python wrapper library based on the zbar library for barcode/QR code detection and decoding
Visual Tracking	Detects and continuously locks onto a target through a vision algorithm; only the pan-tilt follows, the car does not move
PTZ Pan-Tilt	A two-degree-of-freedom servo pan-tilt; s1 rotates horizontally and s2 tilts, controlled via the `/cmd_ptz_angle` topic
PD Control	Proportional-derivative control; computes the pan-tilt angle step from the target offset and the offset change rate
Deadband	When the target offset is within the deadband pixel range, the pan-tilt does not move, avoiding continuous jitter caused by tiny errors

2. Technical Architecture

The QR code tracking node `qr_tracker_ptz_node` subscribes to the raw image topic `/camera/image_raw` of the ESP32 camera, detects QR codes with the pyzbar or OpenCV dual backend, parses the payload, computes the pan-tilt angle step with a PD controller, and publishes to `/cmd_ptz_angle` to control the pan-tilt to track the QR code.

Detection pipeline: the `Image` message is converted via `CvBridge.imgmsg_to_cv2` → scaled to 640×480 → orientation correction → QR code detection (pyzbar/opencv/auto) → multi-code selection strategy (largest area / nearest center / first) → target center EMA filter → PD pan-tilt control (fixed 20 Hz).

3. Core Topics Quick Reference

Firmware Uplink (Sensor Data)

Topic	Message Type	QoS	Description
/camera/image_raw	sensor_msgs/msg/Image	best_effort	Raw image from the ESP32 camera

Firmware Downlink (Control Commands)

Topic	Message Type	QoS	Field	Range	Direction
/cmd_ptz_angle	geometry_msgs/msg/Vector3	best_effort + keep_last(1)	x	[-90, 90]°	s1 horizontal servo angle
			y	[-90, 20]°	s2 tilt servo angle

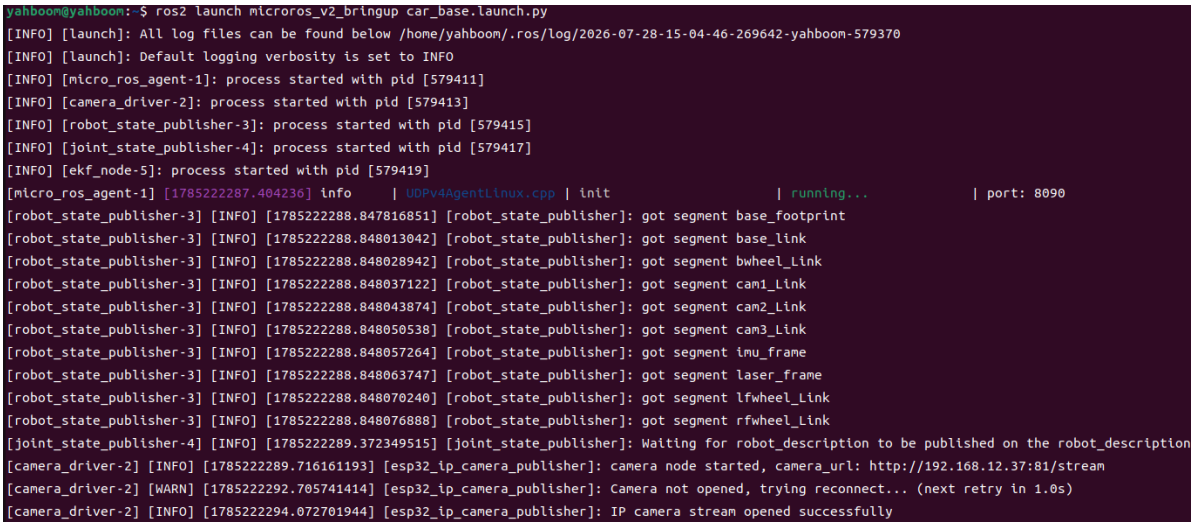
3. Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ version	☑ Supported
No-PTZ version	✗
Required peripherals	ESP32 camera (pan-tilt) + PTZ servo

2. Start Agent + Camera + Joint Model (Terminal 1)

```
ros2 launch microros_v2_bringup car_base.launch.py
```



3. Start QR Code Tracking (Terminal 2)

```
ros2 launch microros_v2_ptz_visual_control_pkg qr_tracker_ptz.launch.py
```

4. Operation Instructions

Workflow

After startup, the node runs with two independent timers (image processing at 30 Hz + pan-tilt control at 20 Hz):

1. Image decoding and preprocessing

The image callback converts and caches the raw image; the image processing timer takes the latest frame, scales it to 640×480, applies orientation correction, and enters the detection pipeline.

2. QR code detection

Dual-backend detection is supported:

- **pyzbar (default)**: grayscale image → `pyzbar.decode` scan, extracting the payload text, rectangle, and polygon vertices
- **opencv**: `cv2.QRCodeDetector.detectAndDecode` / `detectAndDecodeMulti`, supporting simultaneous detection of multiple codes
- **auto**: tries pyzbar first, falls back to opencv when it returns nothing

3. Multi-code selection strategy

If multiple QR codes are detected at the same time, the primary target is chosen by `qr_select_strategy`:

- `largest` (default): selects the code with the largest area (rect.width × rect.height)
- `nearest_center`: selects the code closest to the frame center
- `first`: selects the first detected code

4. Target state update

When a QR code is detected, the target state is updated (payload, rectangle, polygon vertices, center coordinates) and the update time is recorded. If no code is detected for more than `lost_timeout` (0.5 s), the target is marked lost.

5. PD pan-tilt control (tracking state only)

Fixed-rate 20 Hz PD control: uses the QR code center as the tracking point, normalizes the pixel error → PD computation → step limiting (±1.2°) → step smoothing (α=0.35) → direction reversal (`ptz_x_reverse=true`, `ptz_y_reverse=true`) → angle update → publish to `/cmd_ptz_angle`.

6. Decoded content display

The QR code payload is displayed in the on-screen overlay in real time (truncated with an ellipsis beyond 36 characters); the full content can also be seen in the node log.

Interaction

Key	Function	Description
Space	Tracking/pause switch	Toggles the pan-tilt control; QR codes are still detected and displayed while paused
i / I	Recognition mode	Only detects and decodes, without controlling the pan-tilt
r / R	Reset target	Clears the current QR target, stops control, and returns to recognition mode
q / Q / ESC	Exit	Force-exits the node process with `os._exit(0)`

On-screen overlay: The debug window shows the current state (tracking/identify/lost/paused), frame rate, QR payload (truncated), target age, pan-tilt angle, and PD parameters. A yellow crosshair is drawn at the frame center; detected QR codes are drawn with a green polygon + red center point + blue payload label.

Safety and Edge Cases

- **pyzbar unavailable**: When pyzbar is not installed (`import pyzbar` fails), the node degrades to the opencv-only backend automatically and logs a warning at the `debug_log_interval_sec` interval
- **Target lost handling**: When no QR code is detected for more than `lost_timeout` (0.5 s) → the target is marked lost and the PD state resets; if `control_enabled=true`, the node switches to the `lost` state
- **Pan-tilt angle limiting**: Horizontal [-90°, 90°], tilt [-90°, 20°]
- **Deadband protection**: The step is forced to zero when the QR center offset is within the deadband (18 px)
- **Initial publish at startup**: With `publish_initial_ptz_command=true`, the initial pan-tilt angle (0°, -45°) is published repeatedly 20 times at 0.2 s intervals
- **Auto start**: With `auto_start_control=true` (default), the node enters tracking mode automatically as soon as a QR code is detected after startup
- **Exit strategy**: Press q/ESC to exit directly with `os._exit(0)`

5. How to Stop

Press `Ctrl+C` to stop Terminal 2 → Terminal 1 in order.

4. Experiment Observations

- **No QR code**: The frame streams in real time with the state `lost` or `identify`; the pan-tilt stays at its current position
- **QR code detected + tracking mode**: The pan-tilt follows the QR code smoothly; the frame overlays a green polygon + red center point, the payload is shown in real time, and the pan-tilt angle and PD parameters are overlaid
- **QR code detected + recognition mode**: QR code detection and decoding work normally, but the pan-tilt does not move
- **QR code within the deadband**: The step is zeroed and the pan-tilt stays still without jitter
- **QR code lost by timeout (>0.5 s)**: Enters the `lost` state, the PD state resets, and the pan-tilt stays at its last position
- **Multiple QR codes**: All codes are drawn with green polygons; the primary target code (selected by the strategy) is highlighted with a purple polygon, and the IDs of multiple codes are shown on the frame
- **Decoding failure (content cannot be parsed)**: The QR position is tracked normally, but the payload is shown as empty

5. Program Analysis

> Source code paths:
>
> - Launch: `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/launch/qr_tracker_ptz.launch.py`
> - Node: `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/microros_v2_ptz_visual_control_pkg/qr_tracker_ptz_node.py`

1. Launch Parameters

> Parameter definition files:
>
> - Launch: `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/launch/qr_tracker_ptz.launch.py`
> - Node: `~/microros_ws/src/microros_v2_ptz_visual_control_pkg/microros_v2_ptz_visual_control_pkg/qr_tracker_ptz_node.py`

Parameter	Type	Default Value	Description
`image_topic`	string	`/camera/image_raw`	Raw image input topic
`ptz_command_topic`	string	`/cmd_ptz_angle`	Pan-tilt angle output topic
`window_name`	string	`qr_control_ptz`	OpenCV debug window name
`show_debug_window`	bool	`true`	Whether to show the debug window
`qr_detector_backend`	string	`pyzbar`	Detection backend: pyzbar / opencv / auto
`qr_select_strategy`	string	`largest`	Multi-code selection strategy: largest / nearest_center / first
`control_rate_hz`	float	`20.0` Hz	Pan-tilt control publishing rate
`image_processing_rate_hz`	float	`30.0` Hz	Image processing rate
`auto_start_control`	bool	`true`	Start tracking automatically once a QR code is detected after launch
`lost_timeout`	float	`0.5` s	Target lost timeout
`publish_initial_ptz_command`	bool	`true`	Whether to publish the initial pan-tilt angle at startup
`initial_horizontal_deg` / `initial_pitch_deg`	float	`0.0°` / `-45.0°`	Initial pan-tilt angles
`pd_kp` / `pd_kd`	float	`0.7` / `0.25`	PD proportional/derivative gains
`ptz_step_smooth_alpha`	float	`0.35`	Step smoothing coefficient
`max_ptz_step`	float	`1.2°`	Maximum single angle step
`deadband_x` / `deadband_y`	float	`18.0` / `18.0` px	Horizontal/vertical deadband
`ptz_x_min` / `ptz_x_max`	float	`-90.0°` / `90.0°`	Horizontal angle range
`ptz_y_min` / `ptz_y_max`	float	`-90.0°` / `20.0°`	Tilt angle range
`ptz_x_reverse` / `ptz_y_reverse`	bool	`true` / `true`	Horizontal/tilt direction reversal
`debug_log_interval_sec`	float	`2.0` s	Debug log interval (0 disables)
`enable_rqt_dynamic_parameters`	bool	`true`	Allow runtime dynamic tuning via rqt

2. Core Node Analysis

`QrControlPtzNode` inherits from `rclpy.node.Node` and is the core implementation of QR code tracking. The constructor and the PD control callback are shown below.

Constructor `__init__`

The constructor declares the ROS2 parameters, creates the publishers and subscribers, initializes the tracking state, and sets up the dual timers for image processing and pan-tilt control:

```
class QrControlPtzNode(Node):
    """QR code PTZ tracking node."""

    def __init__(self) -> None:
        super().__init__("qr_tracker_ptz_node")

        self.image_lock = threading.Lock()
        self.target_lock = threading.Lock()

        self.qr_detector = cv2.QRCodeDetector()
        self.latest_display_frame = None
        self.running = True

        self._declare_qr_tracker_parameters()
        self.load_parameters()

        # Initial pan-tilt angles
        self.current_horizontal_deg = clamp(
            self.initial_horizontal_deg, self.ptz_x_min, self.ptz_x_max
        )
        self.current_pitch_deg = clamp(
            self.initial_pitch_deg, self.ptz_y_min, self.ptz_y_max
        )
        self.control_enabled = self.auto_start_control
        self.state = "tracking" if self.control_enabled else "identify"

        # Create the image subscriber (raw image)
        self.image_subscription = self.create_subscription(
            Image,
            self.image_topic,
            self.image_callback,
            QoSProfile(history=QoSHistoryPolicy.KEEP_LAST, depth=1,
                       reliability=QoSReliabilityPolicy.BEST_EFFORT,
                       durability=QoSDurabilityPolicy.VOLATILE),
        )

        # Create the pan-tilt angle publisher
        self.ptz_publisher = self.create_publisher(
            Vector3, self.ptz_command_topic, 10
        )

        # Create the image processing timer (30 Hz by default)
        self.image_timer = self.create_timer(
            1.0 / max(self.image_processing_rate_hz, MIN_TIMER_RATE_HZ),
            self.process_latest_image,
        )
```

```
        # Create the pan-tilt control timer (20 Hz by default)
        self.control_timer = self.create_timer(
            1.0 / max(self.control_rate_hz, MIN_TIMER_RATE_HZ),
            self.update_ptz_tracking,
        )

        # Start repeated publishing of the initial pan-tilt angle
        if self.publish_initial_ptz_command:
            self.initial_timer = self.create_timer(
                max(self.initial_publish_interval_sec, 0.02),
                self.publish_initial_ptz,
            )

        # Register the runtime dynamic tuning callback
        self.parameter_callback = self.add_on_set_parameters_callback(
            self.on_parameter_update
        )
```

PD control callback `update_ptz_tracking`

Triggered at a fixed rate (20 Hz); computes the QR center offset and runs PD pan-tilt control:

```
def update_ptz_tracking(self) -> None:
    if not self.running or not self.control_enabled:
        return

    with self.target_lock:
        target = self.qr_target
        target_age = time.perf_counter() - self.last_target_time

    # Lost detection
    if (not target.found or target_age > self.lost_timeout
            or self.image_width <= 0 or self.image_height <= 0):
        self.state = "lost"
        self.reset_pd_state()
        return

    # Time delta
    now = time.perf_counter()
    dt = max(now - self.previous_control_time, 1.0e-3)

    # Compute the pixel error (zeroed in the deadband)
    image_cx = self.image_width * 0.5
    image_cy = self.image_height * 0.5
    err_x = target.center[0] - image_cx
    err_y = target.center[1] - image_cy
    if abs(err_x) < self.deadband_x:
        err_x = 0.0
    if abs(err_y) < self.deadband_y:
        err_y = 0.0

    # Normalization + derivative + PD
    norm_x = err_x / image_cx
    norm_y = err_y / image_cy
    dx = (norm_x - self.previous_error_x) / dt
    dy = (norm_y - self.previous_error_y) / dt
    raw_x = (self.pd_kp * norm_x + self.pd_kd * dx) * self.max_ptz_step
```

```
    raw_y = (self.pd_kp * norm_y + self.pd_kd * dy) * self.max_ptz_step
    raw_x = clamp(raw_x, -self.max_ptz_step, self.max_ptz_step)
    raw_y = clamp(raw_y, -self.max_ptz_step, self.max_ptz_step)

    # Step EMA smoothing
    alpha = self.ptz_step_smooth_alpha
    step_x = (1.0 - alpha) * self.last_step_x + alpha * raw_x
    step_y = (1.0 - alpha) * self.last_step_y + alpha * raw_y

    # Save the control history
    self.previous_error_x = norm_x
    self.previous_error_y = norm_y
    self.previous_control_time = now
    self.last_step_x = step_x
    self.last_step_y = step_y

    # Accumulate the angle after direction reversal
    self.current_horizontal_deg += step_x if self.ptz_x_reverse else -step_x
    self.current_pitch_deg += -step_y if self.ptz_y_reverse else step_y

    # Clamp the angles and publish
    self.current_horizontal_deg = clamp(
        self.current_horizontal_deg, self.ptz_x_min, self.ptz_x_max
    )
    self.current_pitch_deg = clamp(
        self.current_pitch_deg, self.ptz_y_min, self.ptz_y_max
    )
    self.publish_ptz_angle(
        self.current_horizontal_deg, self.current_pitch_deg
    )
```

6. Frequently Asked Questions

Problem	Cause	Solution
Blur caused by decoding + tracking conflict	The QR code moves too fast, producing motion blur on the camera	Slow down the QR code movement
QR code not detected	pyzbar not installed or insufficient light	Install `python3-pyzbar` + `libzbar0`, improve the lighting
pyzbar error	Missing dependency libraries	Switch to the opencv backend: `ros2 param set /qr_tracker_ptz_node qr_detector_backend opencv`
Tracking position offset	PD response lags when the QR code moves fast	Increase `pd_kp` (e.g., 1.0), reduce `ptz_step_smooth_alpha` (e.g., 0.2)